

Closed-Loop Bayesian Optimization of Multi-Material 3D-Printed Tensegrity-Inspired Energy Absorbers

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Track: DAC-10 — Design of Engineering Materials and Structures

Tensegrity-inspired architectures, in which rigid struts are suspended within a continuous flexible network, can exhibit tunable nonlinear mechanical responses and ~~favorable~~ high strength-to-weight ~~characteristics~~ ratios, motivating their use in lightweight protective and energy-absorbing structures. Multi-material fused deposition modeling (FDM) can ~~now~~ co-print rigid (PLA) struts and flexible (TPU) elements in a single build, but the resulting design space—spanning strut geometry, tension-element cross-section, connectivity topology, and unit-cell tiling—is too large ~~for efficient to explore~~ efficiently by trial-and-error experimentation.

This work presents a closed-loop experimental campaign ~~guided by that uses~~ Bayesian optimization (BO) ~~for developing to develop~~ multi-material 3D-printed tensegrity-inspired structures ~~optimized for~~ for improved energy absorption under quasi-static compression and drop-weight impact loading. The key contribution is a design–print–test ~~loop that operates directly on physical measurements~~ workflow in which experimental measurements are used directly to update a Gaussian-process surrogate model, enabling sequential design selection without relying on finite-element simulation: ~~a Gaussian-process surrogate model. The surrogate is trained on experimental measurements of~~ peak transmitted force, specific energy absorption (SEA), and compaction efficiency, and ~~an a batch~~ acquisition function recommends the next ~~batch of set of candidate~~ designs after each test round. The workflow ~~is largely automated—design generation, surrogate updating, and acquisition-function optimization proceed without operator input, while specimen loading and fixture placement remain the primary manual steps~~ combines automated model updating and experiment selection with manual specimen handling and test setup.

We define a parameterized family of unit cells using a core-wrapping ~~strategy~~ architecture in which rigid PLA struts are enclosed by continuous TPU skins to improve interfacial interlocking. Design variables include strut diameter and length, tension-element width and thickness, ~~number of struts~~ strut count per unit cell, and tiling pattern. Specimens are fabricated on a ~~consumer-grade~~ multi-material FDM system and tested under quasi-static compression and instrumented drop-weight impact.

We will report ~~the progression of how~~ BO-selected designs evolve across iterations, ~~the evolution of how~~ the surrogate model changes as data accumulate, and what Pareto-efficient trade-offs emerge between SEA and peak transmitted force within the explored design space. We will also discuss practical lessons for operating ~~a the~~ closed-loop ~~physical experiment~~ workflow, including print-failure handling, batch-to-batch variability, and the exploration–exploitation ~~balance~~ trade-off.